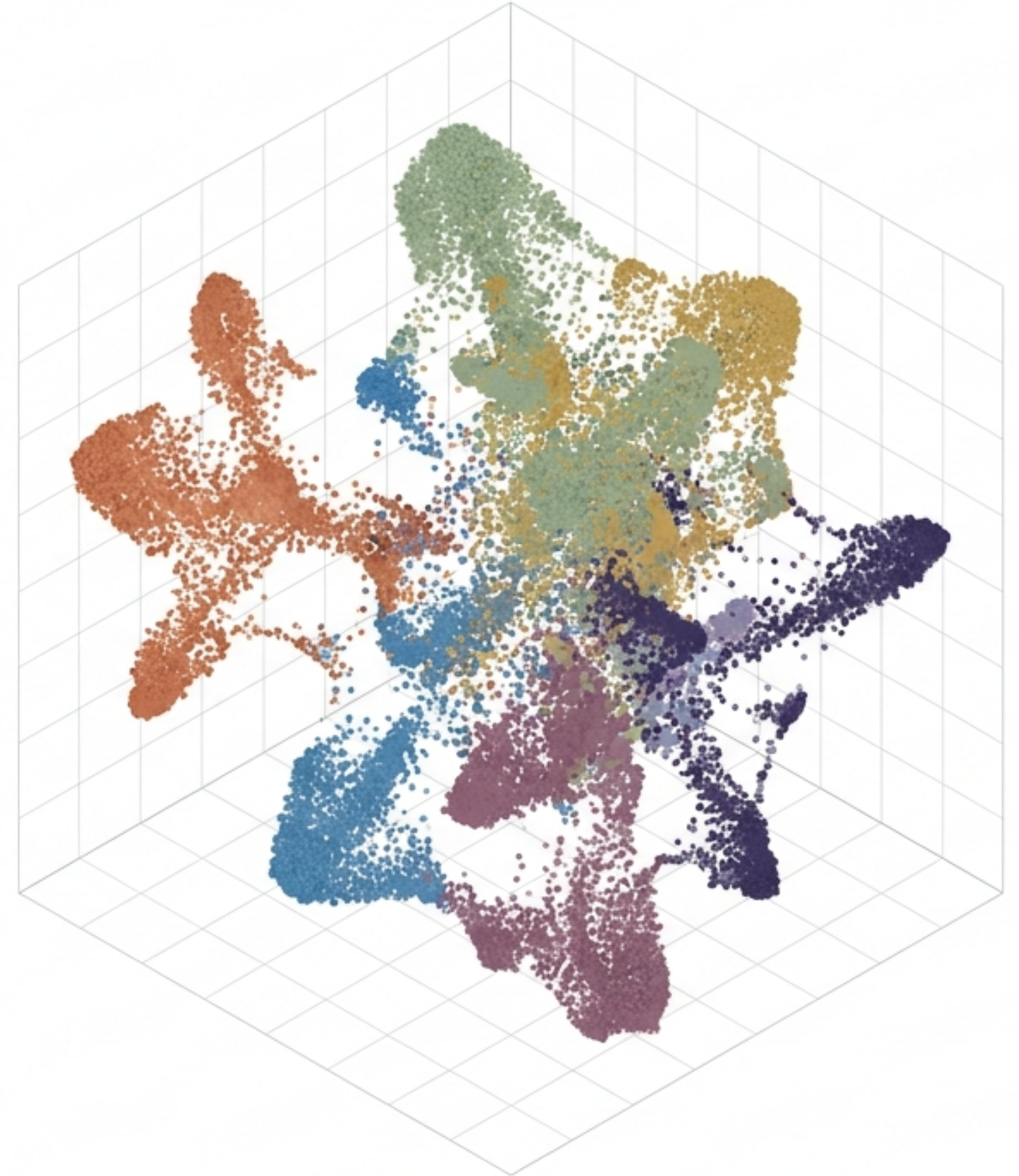


Ground Truth Multi-Modal Brain Transcriptomics

A Reference Architecture

VERSION_1.0 | N=5,000 | 10_MODULES | 14_REGULATORS



Dataset Topology & Relational Schema

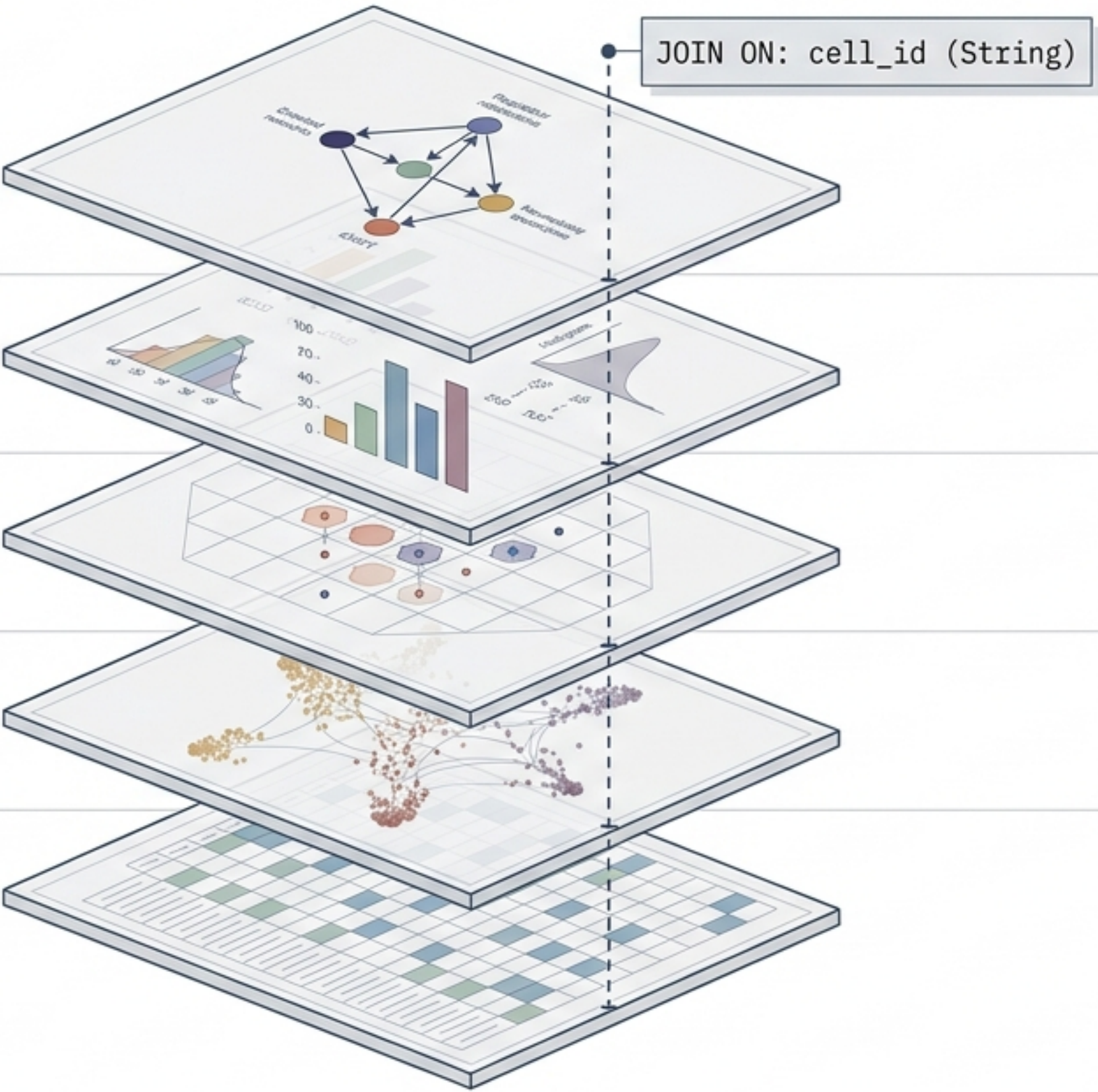
Ground Truth Drivers
ground_truth_regulators.csv

Transcriptomic Statistics
gene_metadata.csv

Physical Geometry
embeddings_spatial.csv

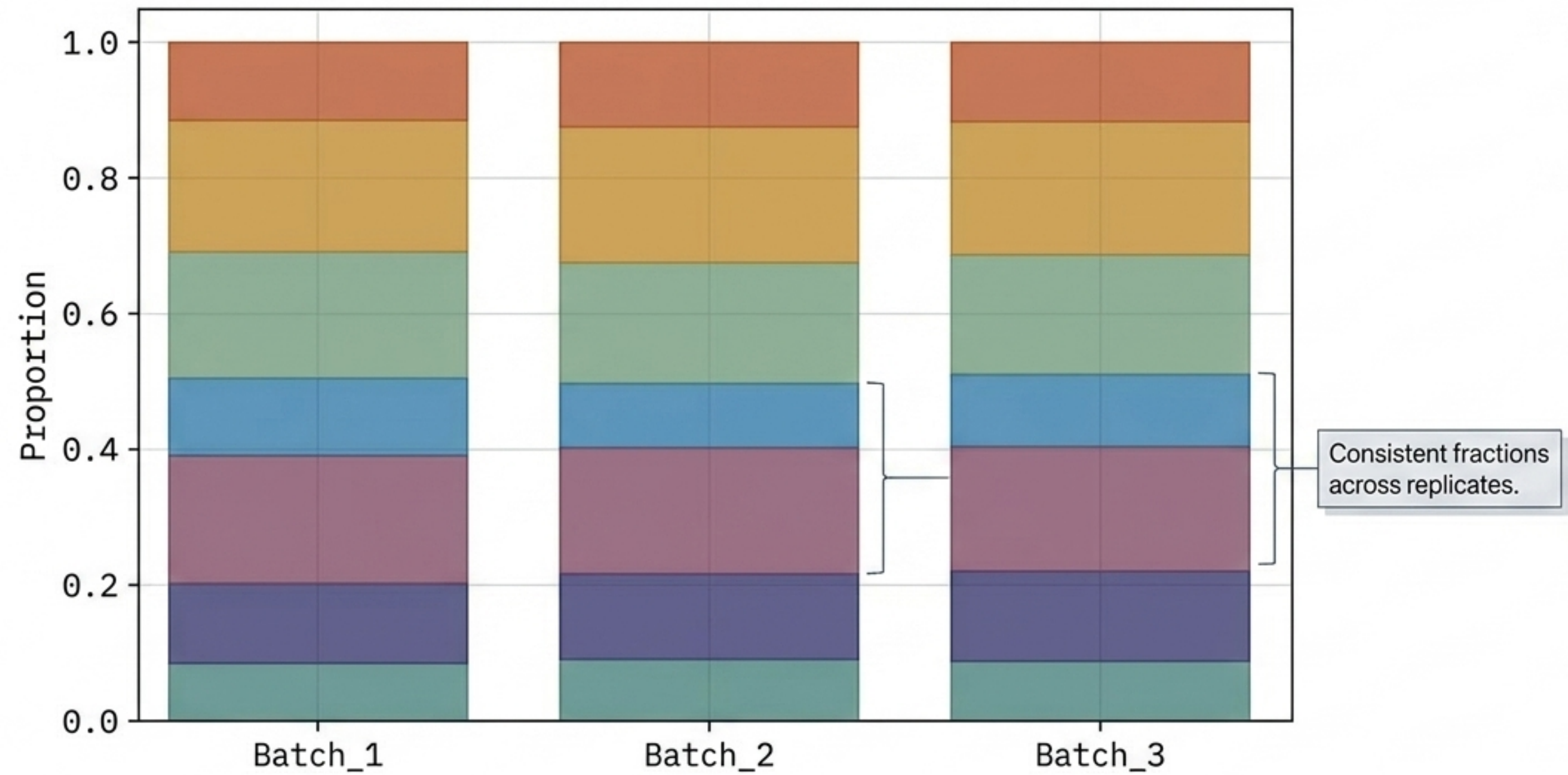
Topological Embeddings
embeddings_umap*.csv

Metadata Matrix
cell_metadata.csv



Cellular Demographics & Batch Distributions

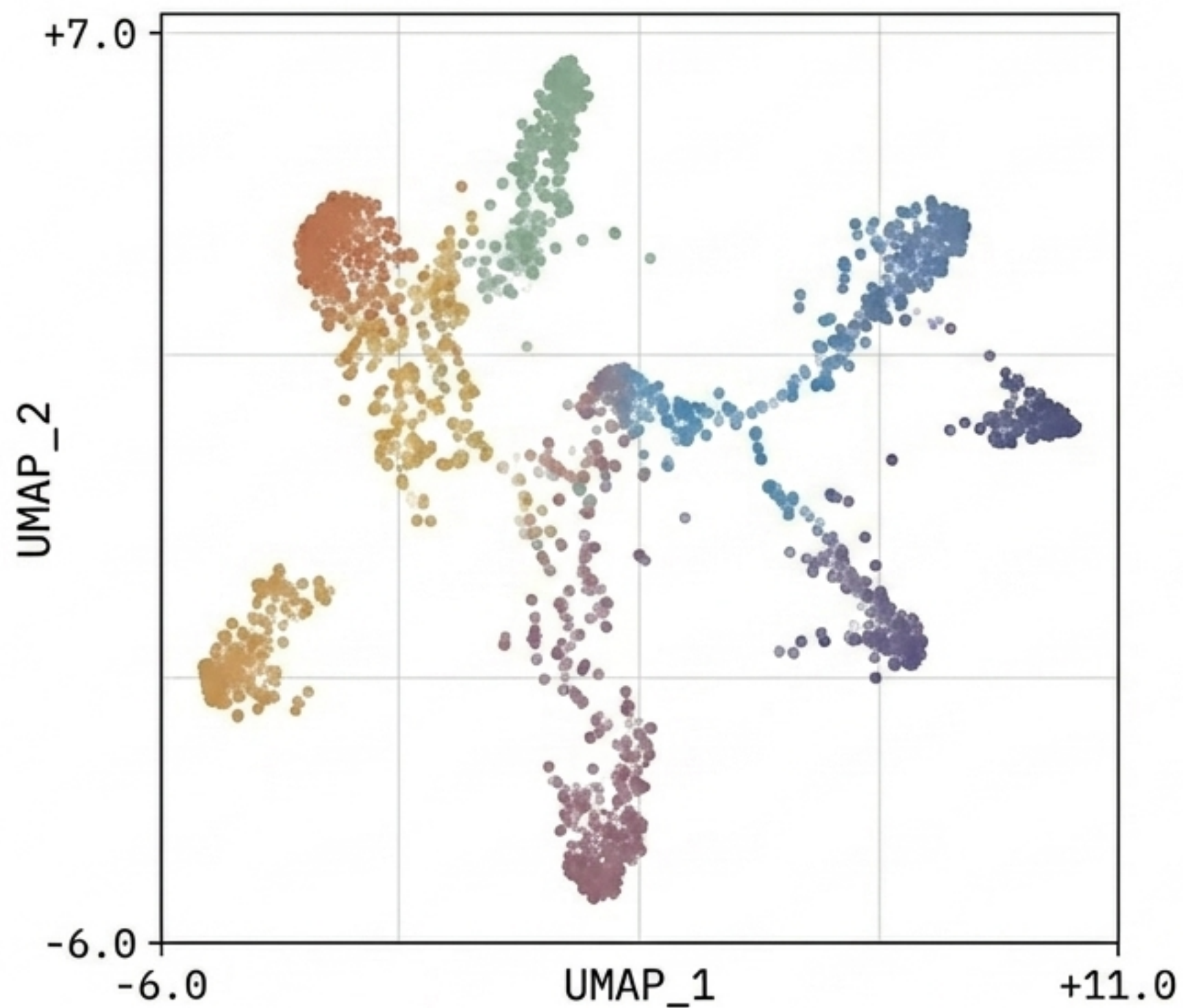
Balanced technical replicates (Batches 1-3) enable robust benchmarking of integration algorithms (e.g., scVI, Seurat CCA).



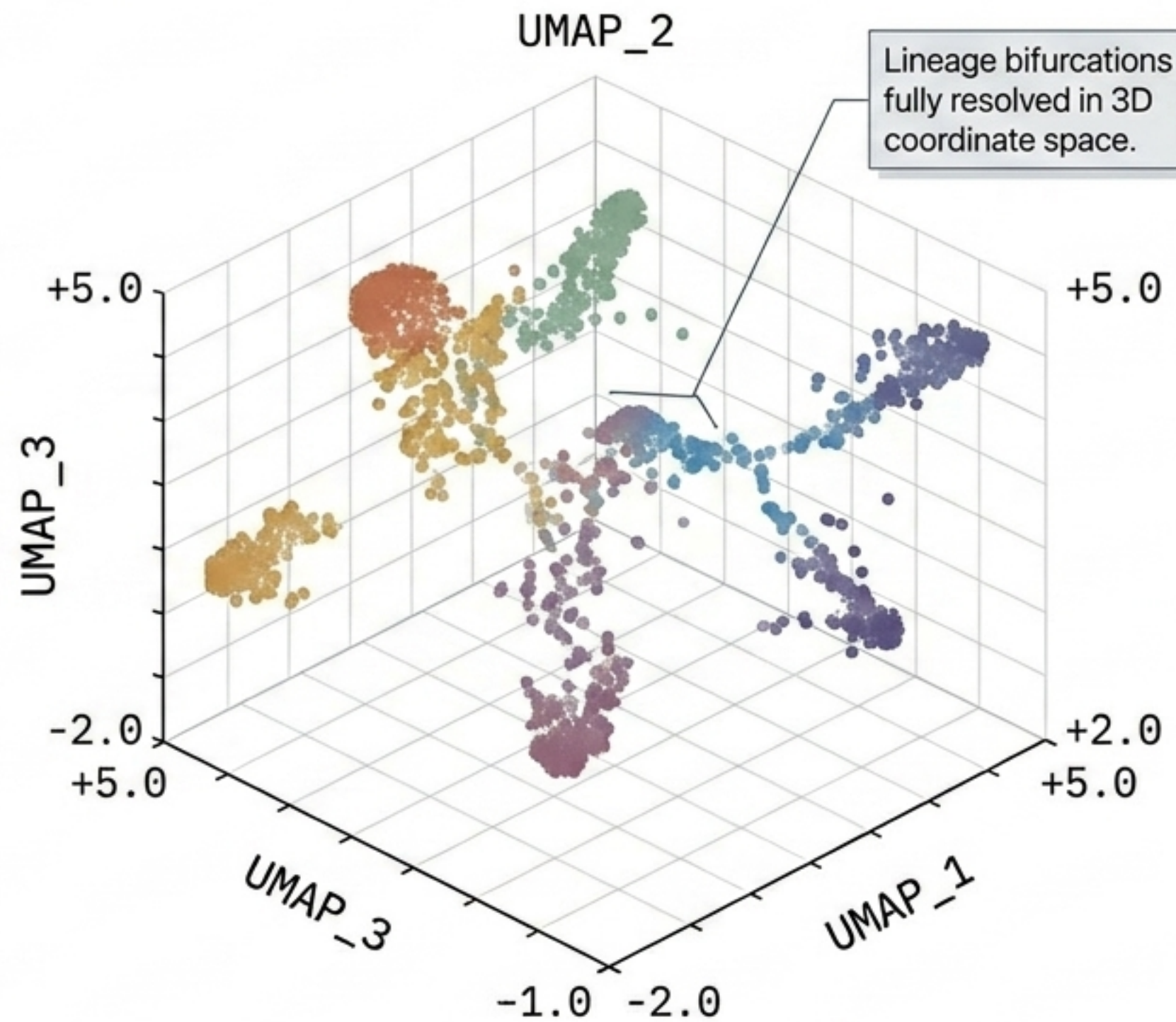
	Astrocyte
	Excitatory_Neuron
	Inhibitory_Neuron
	Microglia
	NPC
	Oligodendrocyte
	OPC

Topological Embeddings: UMAP Space

2D Projection: embeddings_umap2d.csv

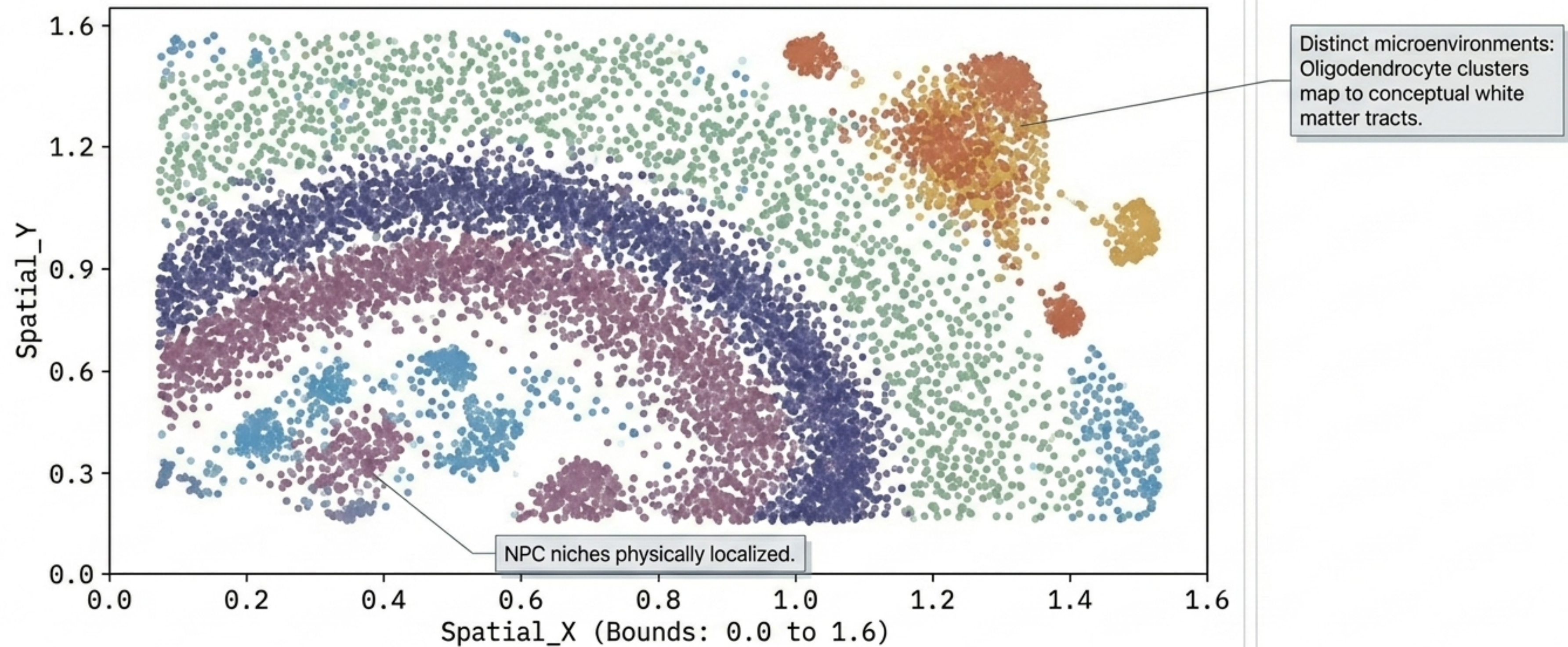


3D Projection: embeddings_umap3d.csv



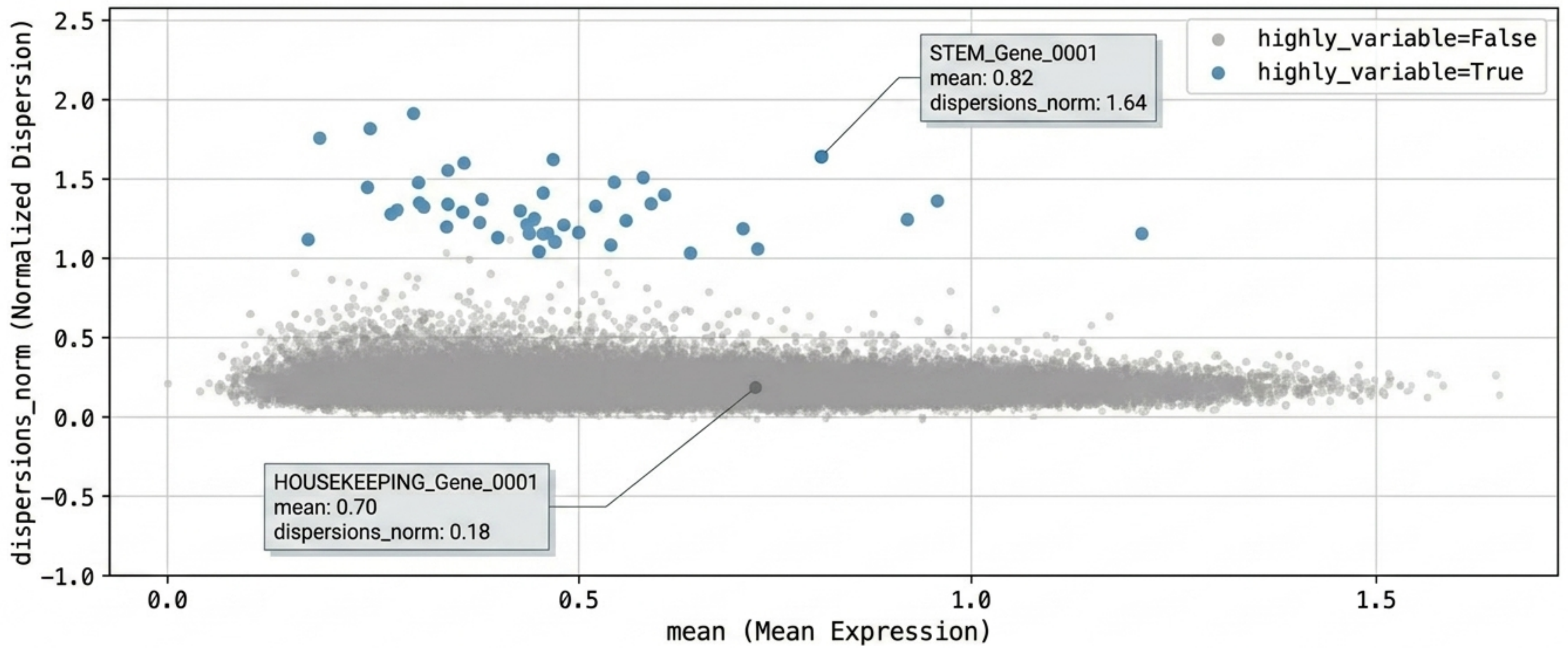
Spatial Architecture: In Situ Coordinate Mapping

Mathematical manifolds translated into physical tissue geometry.

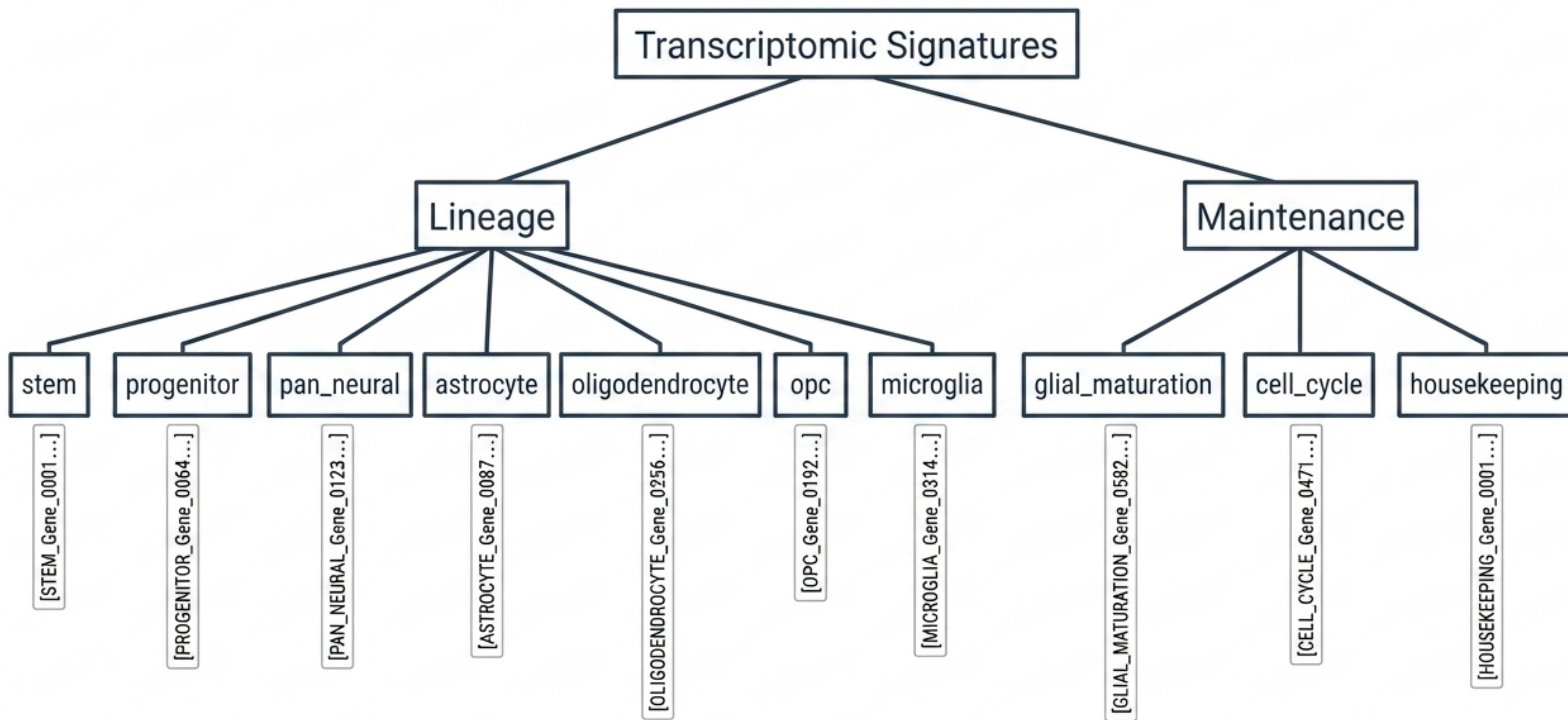


Transcriptomic Parameters: Overdispersion & HVGs

Mathematical isolation of gene modules from background noise.

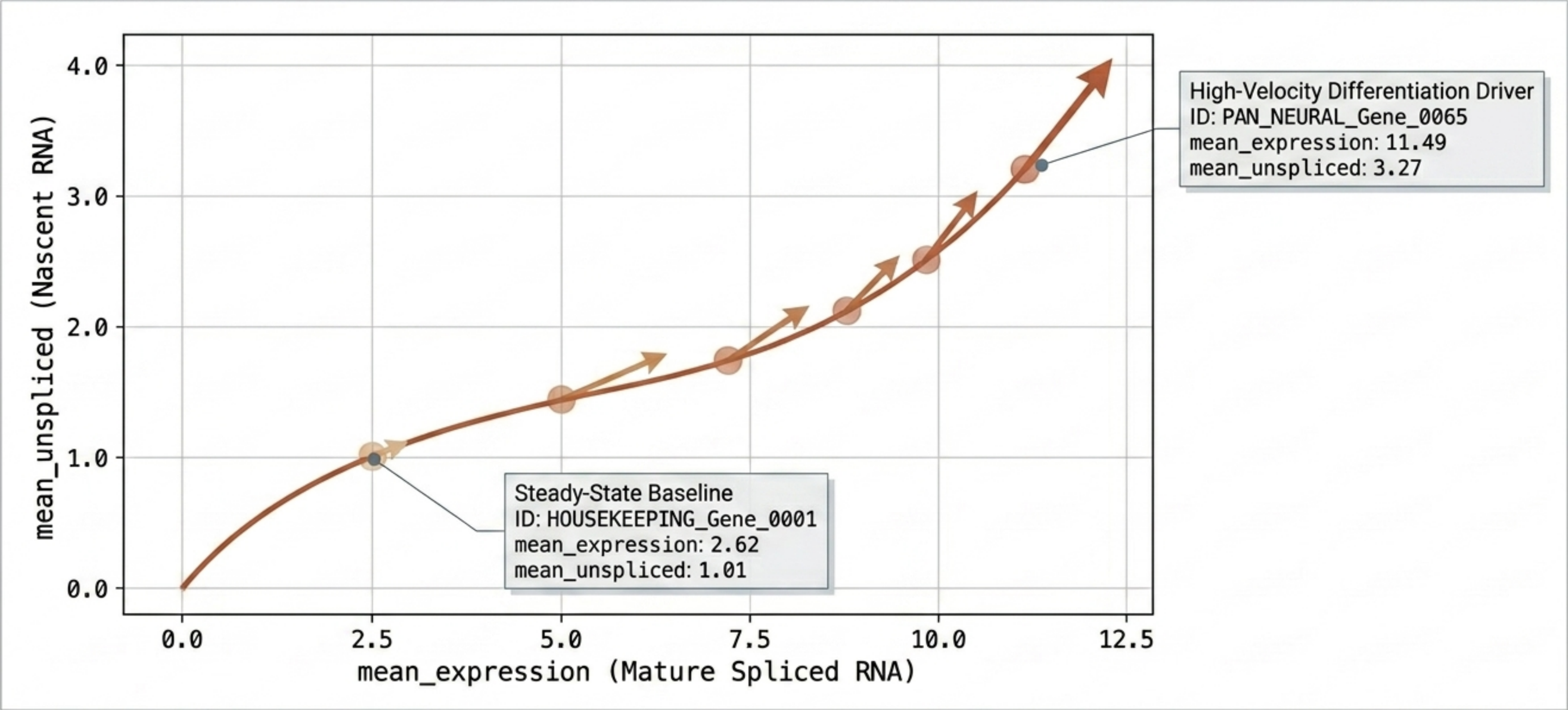


Gene Module Taxonomy



RNA Velocity Indicators: Spliced vs. Unspliced Momentum

Ground-truth momentum vectors enabling precise calibration of stochastic differentiation models.



Ground Truth Regulators: Lineage vs. State



Lineage Initiation

SOX2_like
PAX6_like
NEUROD1_like



Terminal Identity

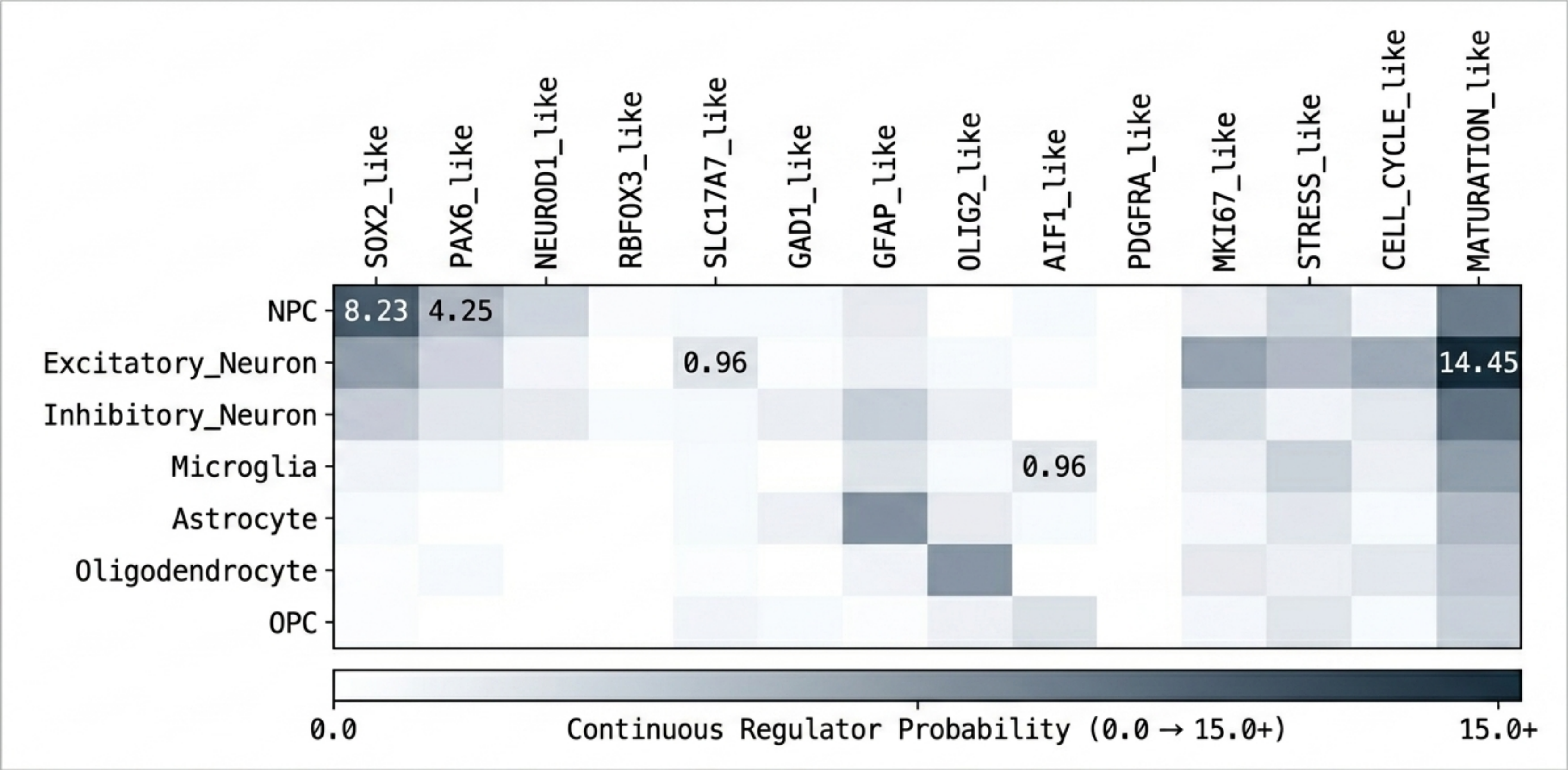
SLC17A7_like (Excitatory)
GAD1_like (Inhibitory)
GFAP_like (Astrocyte)
OLIG2_like (Oligo)
AIF1_like (Microglia)
PDGFRA_like (OPC)
RBF0X3_like (Pan-neuronal)



Cellular State

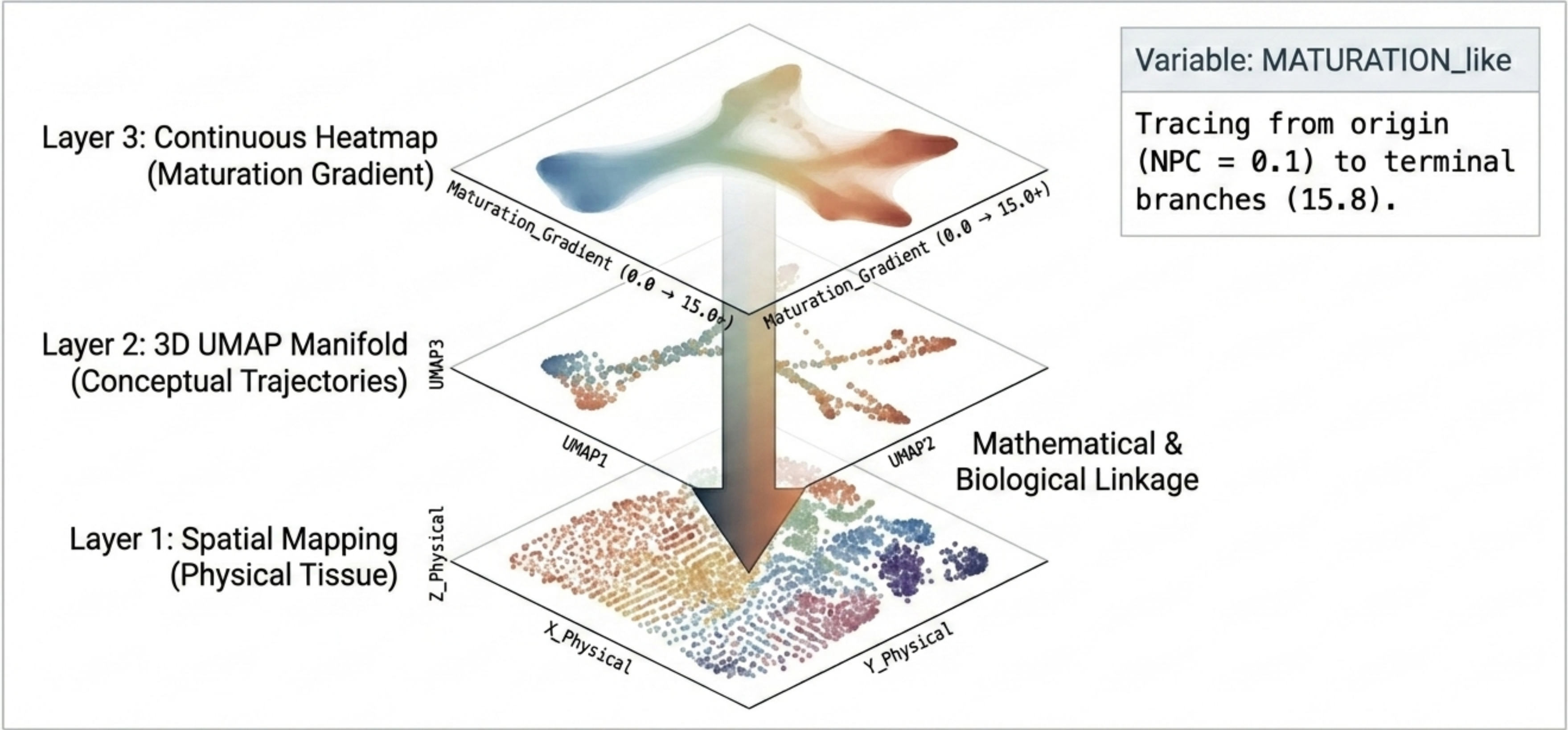
MKI67_like (Proliferation)
STRESS_like
CELL_CYCLE_like
MATURATION_like

Diagnostic Matrix: Cell Type vs. Regulator Drivers



State, Space, and Fate: Trajectory Synthesis

Overlaying continuous gradients onto physical and mathematical manifolds to reveal developmental vectors.



Data Dictionary & Variable Constraints

DataFrame	Key Fields	Data Type	Constraint/Bounds
cell_metadata.csv	Cell Type, Batch	Categorical / String	7 Types, 3 Batches
gene_metadata.csv	highly_variable, mean_expression, dispersions_norm	Bool, Float, Float	Mean > 0.0
ground_truth_regulators.csv	SOX2_like ... MATURATION_like	Float	Continuous Variable
embeddings_spatial.csv	Spatial_X, Spatial_Y	Float32	0.0 to 1.6
embeddings_umap*.csv	UMAP_1, UMAP_2, UMAP_3	Float32	Calculated Manifold